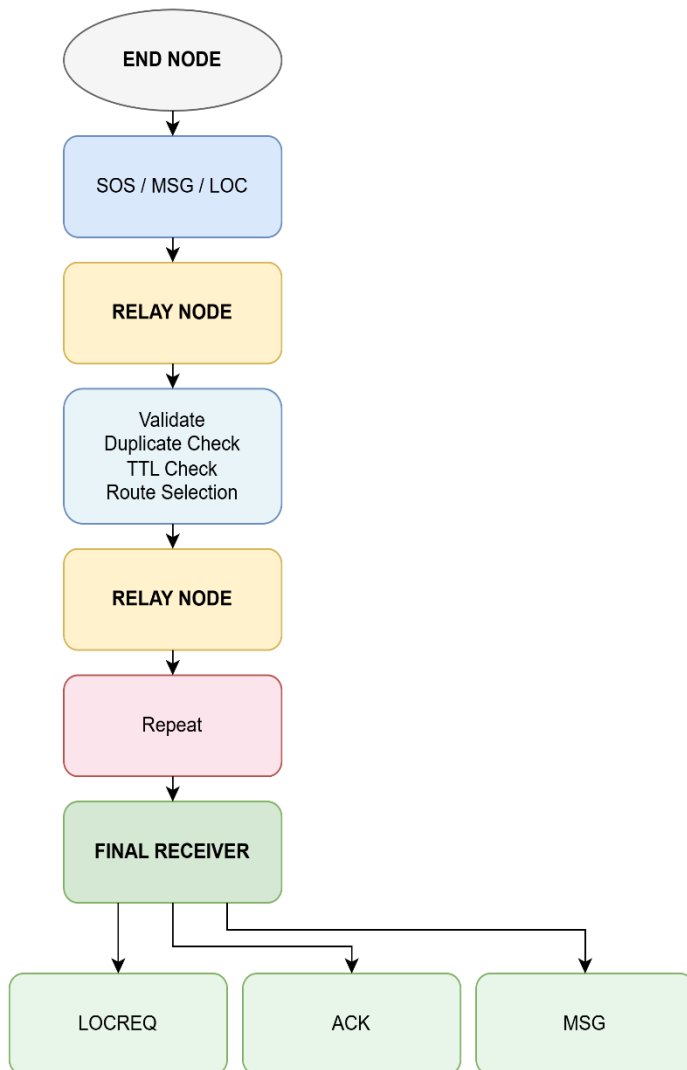


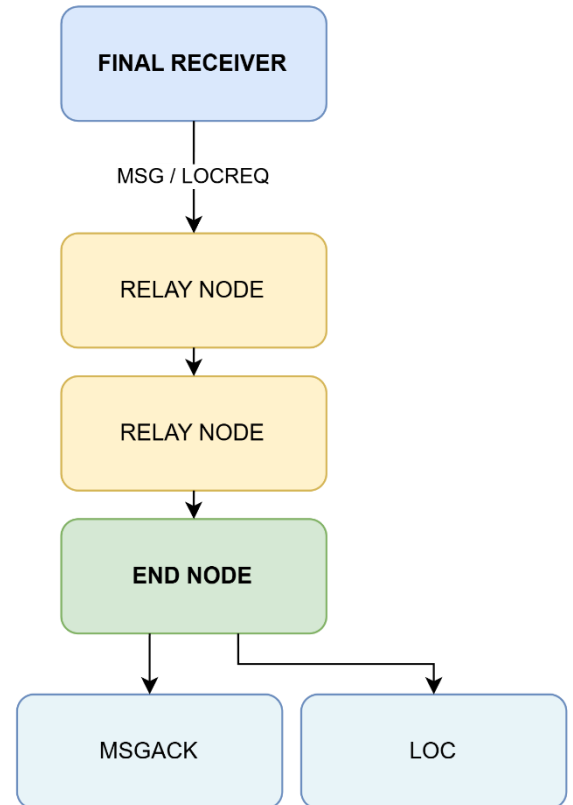
COMMUNICATION PROTOCOL

Communication Model

The latest TerraLink protocol follows a **store-process-forward** communication model.



The **reverse direction** is also supported:



Packet Generation:

When an emergency occurs, the End Node generates an SOS packet.

Example:

TYPE = SOS

SRC = 101

DST = 4

SEQ = 25

TTL = 5
HOP = 0
GPS = Available
BAT = 82%

The End Node then transmits the packet to the surrounding relay network.
The End Node's responsibility ends at **packet generation and transmission**. It does not determine the complete network route.

Packet Structure

Every TerraLink packet follows a structured logical format.

1. Packet Structure

Every TerraLink packet follows a structured logical format. A recommended representation is:

TYPE	PID	SRC	DST	SEQ	TTL	HOP	PAYLOAD
(3B)	(3B)	(3B)	(3B)	(2B)	(1B)	(1B)	(4B)

8 Fields (separated by '|')

Example Packet

Total ASCII Payload Size (Example SOS): ~78 Bytes (excluding separators)

SOS	101	101	4	23	5	2	37.7749,122.4194,1,87,He1p!
TYPE	PID	SRC	DST	SEQ	TTL	HOP	PAYLOAD

• For packets requiring additional information, the payload contains the corresponding application data.

3. Packet Fields

TYPE Identifies the purpose of the packet. (See Type table below for all supported types)	TTL Time-To-Live Remaining forwarding lifetime of a packet. Example: Initial TTL = 5 → TTL = TTL - 1 When TTL <= 0, the packet is discarded.																		
PID Packet Identifier Provides an identifier for the packet instance and helps distinguish individual transmissions.	HOP Hop Information Records the forwarding progression of the packet. Helps in diagnostics and route analysis.																		
SRC Source Node ID Identifies the node that originally generated the packet. Example: SRC = 101	PAYLOAD Contains application-specific information. (See payload examples below)																		
DST Destination Node ID Identifies the intended destination. Example: DST = 4 (final receiver) DST = 255 (broadcast)	Payload Examples <table><tbody><tr><td>SOS SOS</td><td>MSG Text Message</td><td>LOC Location</td></tr><tr><td>• GPS Latitude</td><td></td><td>• Latitude</td></tr><tr><td>• GPS Longitude</td><td></td><td>• Longitude</td></tr><tr><td>• GPS Fix Status</td><td></td><td>• GPS Fix</td></tr><tr><td>• Battery Level</td><td></td><td></td></tr><tr><td>• Emergency Information</td><td></td><td></td></tr></tbody></table>	SOS SOS	MSG Text Message	LOC Location	• GPS Latitude		• Latitude	• GPS Longitude		• Longitude	• GPS Fix Status		• GPS Fix	• Battery Level			• Emergency Information		
SOS SOS	MSG Text Message	LOC Location																	
• GPS Latitude		• Latitude																	
• GPS Longitude		• Longitude																	
• GPS Fix Status		• GPS Fix																	
• Battery Level																			
• Emergency Information																			
SEQ Sequence Number Incremented by the source node for new packets. Used for identification, duplicate suppression, repeated transmission detection and freshness.																			

2. Optimized Packet Architecture & Analysis

Optimized Binary Struct

```
#pragma pack(1)
struct TerraFrame {
    uint8_t type;
    uint8_t pid;
    uint16_t src_id;
    uint16_t dst_id;
    int32_t lat_micdeg;
    ...
};
```

Airtime (ToA) Comparison
(at SF10, 125kHz, 4/5)

650ms	280ms
ASCII String Frame	Optimized Binary Frame

- Optimized Address Storage (e.g., uint16_t)
- Int32 Micro-degrees GPS (Lat/Lon)
- Enum Type Definitions
- Static Byte Buffers (No Heap)

Airtime (ToA) Comparison
(at SF10, 125kHz, 4/5)

650ms	280ms
ASCII String Frame	Optimized Binary Frame

Optimized Logical Packet Structures

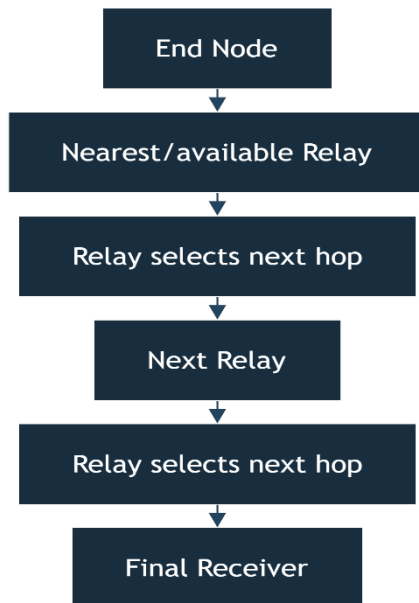
SOS (DATA)	TYPE	PID	DST_ID	PAYLOAD	ROUTE
ACK (CONTROL)	TYPE	PID	DST_ID		
HELLO (NEIGHBOR)	TYPE	PID	SRC_ID	BAT	

Dynamic Route Selection

The latest TerraLink architecture separates **packet generation** from **network routing**.

The End Node does not need to know the complete path to the receiver.

Instead:



Each relay makes a local forwarding decision using its available routing information.

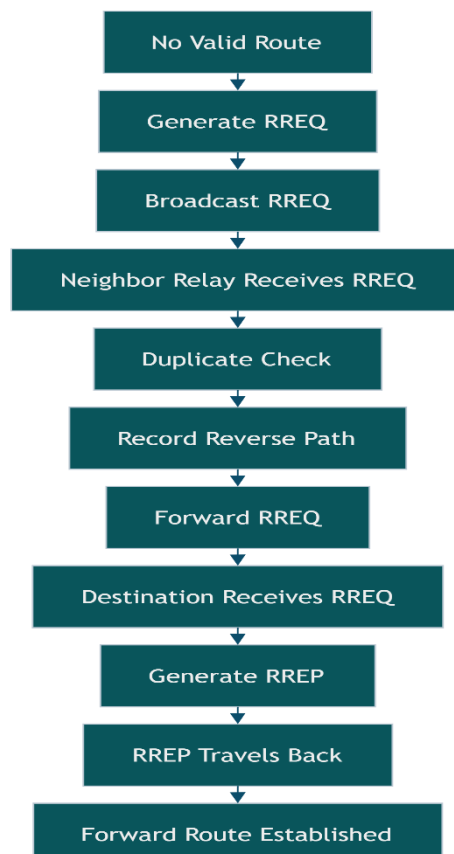
The relay considers:

- Destination
- Available neighbouring nodes
- Known routes
- Link availability
- Route validity
- Previous forwarding information
- Packet TTL

This allows the network to adapt when its

Route Discovery:

The simplified sequence is:



When a relay does not have a valid route toward the destination, TerraLink can initiate a lightweight route-discovery procedure.

The mechanism is **AODV-inspired**, but it is not claimed to be a complete implementation of standard AODV.

RREQ — Route Request

An RREQ is generated when a valid route toward the destination is unavailable.

A relay receiving an RREQ performs:

Step 1 — Packet Validation

Check whether the packet is structurally valid.

Step 2 — Duplicate Check

Check whether the same RREQ has already been processed.

If it is a duplicate:

DISCARD

If it is new:

PROCESS

Step 3 — Reverse Route Recording

The relay records the node from which the RREQ was received.

This provides a reverse path for the RREP.

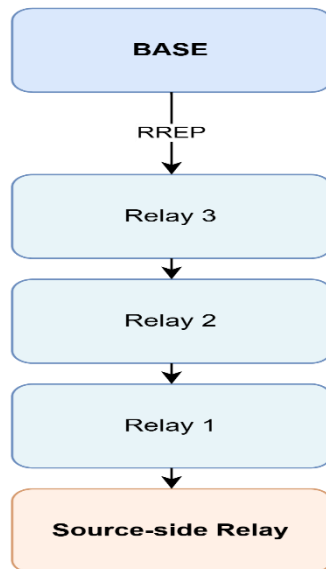
Step 4 — Forward RREQ

The RREQ is forwarded toward other available relay nodes.

RREP — Route Reply

When the destination/base node receives a valid RREQ, it generates an RREP.

The RREP follows the reverse path established during RREQ propagation.



Each relay updates its route information when it receives a valid RREP.

After route establishment:

Source → Relay 1 → Relay 2 → Relay 3 → Base

can be used for subsequent packet forwarding.

Route Information

Each relay maintains lightweight route information.

A logical route entry may contain:

Field	Purpose
Destination	Target node
Next Hop	Immediate forwarding node
Hop Count	Approximate route length
Sequence Information	Route freshness
Validity	Whether route can currently be used

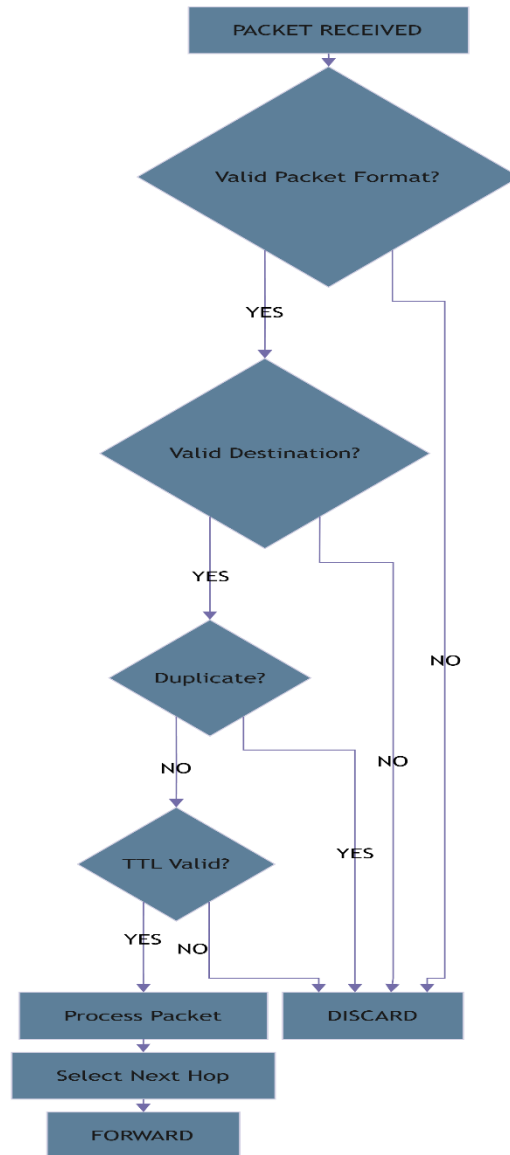
The relay does not need a complete global topology map.

This keeps memory and processing requirements suitable for microcontroller-based nodes.

Packet Validation:

Every relay must validate a received packet before forwarding it.

The processing sequence is:



This prevents invalid packets from consuming network resources.

Duplicate Packet Suppression:

LoRa is a shared broadcast medium, so the same packet may reach multiple relay nodes.

Without duplicate suppression, this could produce excessive forwarding.

TerraLink therefore maintains a small duplicate cache based primarily on:

SRC + SEQ

For example:

SRC = 101

SEQ = 25

If the relay has already processed:

101 + 25

a subsequent copy is discarded.

This prevents:

- Repeated forwarding
- Broadcast storms
- Unnecessary LoRa traffic
- Duplicate SOS processing
- Routing loops

13. TTL-Based Loop Prevention

Every forwarded packet has a finite TTL.

Example:

Initial TTL = 5

Forwarding:

Relay 1 → TTL 4

Relay 2 → TTL 3

Relay 3 → TTL 2

Relay 4 → TTL 1

Relay 5 → TTL 0

Once TTL expires:

PACKET DISCARD

TTL prevents:

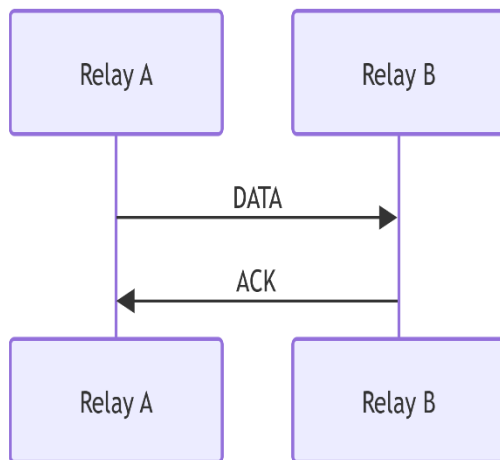
- Infinite packet circulation
- Routing loops
- Excessive channel occupation
- Uncontrolled retransmission

14. Hop-by-Hop Acknowledgement

TerraLink uses acknowledgement at the communication-link level.

Instead of assuming that successful transmission means successful end-to-end delivery, a forwarding node waits for confirmation from the next communication stage.

Conceptually:



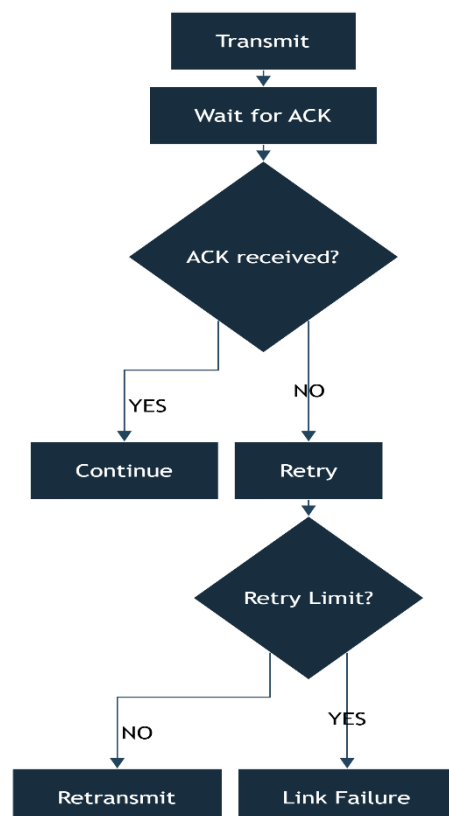
If an ACK is not received within the defined timeout:

ACK TIMEOUT
↓
RETRANSMISSION

A limited number of retries prevents a failed link from blocking the network indefinitely.

15. Retransmission

When a packet is transmitted to a next hop and the expected acknowledgement is not received:



When the retry limit is reached, the relay treats the next-hop link as unavailable.

16. Dynamic Link Failure Detection

A major feature of Dynamic TerraLink is the ability to react to communication failure.

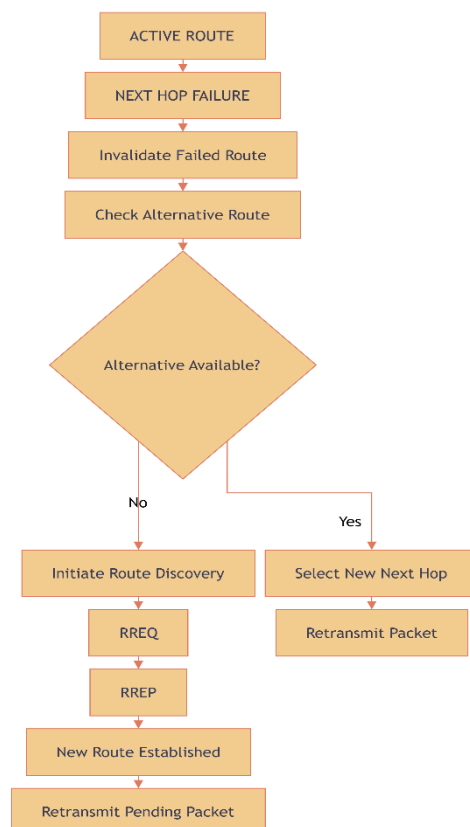
A link may be considered unreliable when:

- ACK is not received
- Multiple retransmissions fail
- The next hop becomes unavailable
- A route becomes invalid
- A relay stops responding

The relay then avoids continuously transmitting through the failed link.

17. Self-Healing Route Recovery

When a next hop fails, TerraLink attempts to recover communication

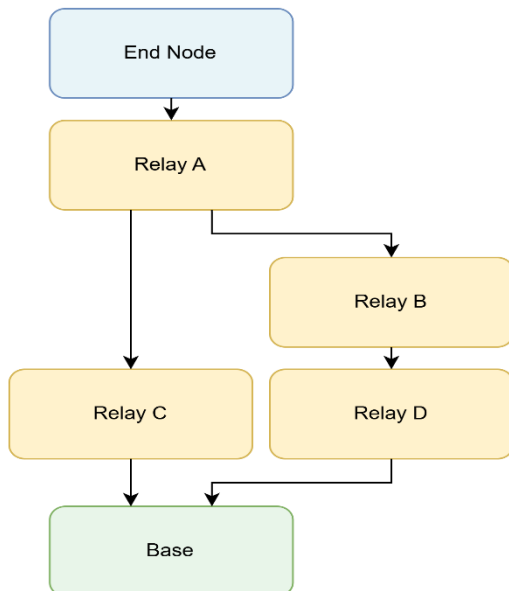


This is the core **dynamic/self-healing mechanism**.

. Alternative Route Selection

If multiple forwarding possibilities exist, the relay can select an available next hop based on its current routing information.

Conceptually:



If:

Relay A → Relay B fails,
the network can attempt:
Relay A → Relay C

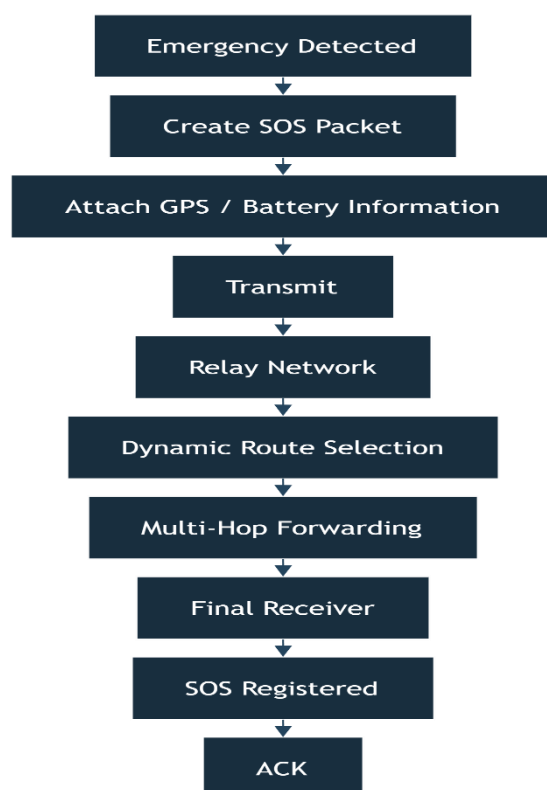
instead of stopping communication
completely.

This is what distinguishes **Dynamic TerraLink**
from a fixed relay chain.

19. SOS Communication

SOS is the highest-priority application packet.

The sequence is:



The command center can then display the emergency incident and associated location information.

20. Bidirectional MSG Communication

The latest protocol is not limited to victim-to-base SOS communication. The Final Receiver can send a MSG packet to a specific End Node.

Example:

SRC = 4

DST = 101

TYPE = MSG

PAYLOAD = "Rescue team is approaching"

The relay network forwards the packet toward node 101.

The destination End Node:

1. Validates the packet.
2. Checks the destination.
3. Suppresses duplicates if already processed.
4. Displays the message.
5. Produces an audible notification.
6. Sends MSGACK back toward the receiver.

21. MSG Handling During Active SOS

A critical latest-protocol requirement is that a node with an active SOS must still be able to receive an operator message.

Therefore, incoming:

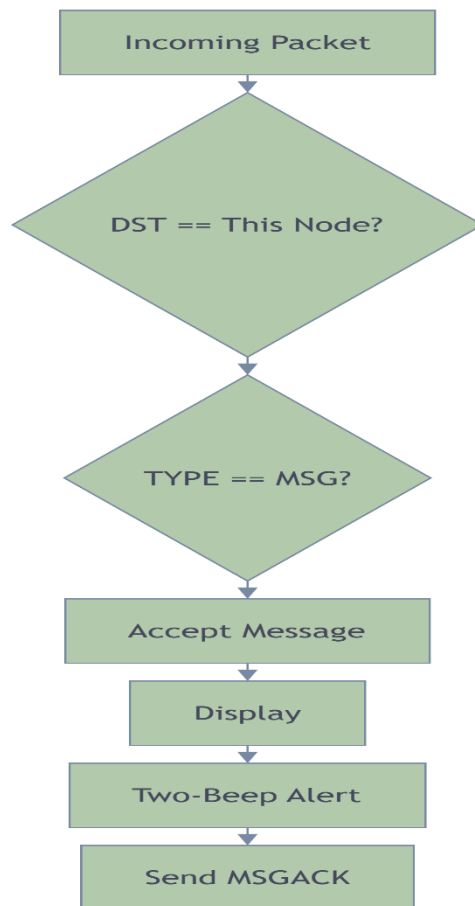
SRC = Receiver

DST = Active SOS Node

TYPE = MSG

must be accepted even if its sequence number does not belong to the active SOS transaction.

The logic is:



This prevents an active emergency state from blocking command-center communication.

22. MSGACK

After successfully receiving an operator message, the End Node sends:

TYPE = MSGACK

SRC = End Node

DST = Receiver

The command center can therefore determine whether the message reached the intended node.

This provides message-delivery status at the application level.

23. Location Request

The Final Receiver can request the current location of a node using:

TYPE = LOCREQ

Example:

SRC = 4

DST = 101

TYPE = LOCREQ

The request travels through the relay network.

The destination End Node responds with:

TYPE = LOC

SRC = 101

DST = 4

GPS = Latitude / Longitude

This provides a bidirectional location mechanism.

24. Packet-Type Processing:

The latest protocol can be summarized as follows:

Packet	Origin	Destination	Main Function
SOS	End Node	Base	Emergency alert
ACK	Receiver/Relay	Previous node	Delivery confirmation
MSG	Receiver/Operator	End Node	Command/message
MSGACK	End Node	Receiver	Message confirmation
LOCREQ	Receiver	End Node	Location request
LOC	End Node	Receiver	Location response
RREQ	Routing node	Network	Route discovery
RREP	Destination	Routing node	Route establishment
RERR	Relay	Relevant nodes	Route failure notification

27. Reliability Mechanisms

Dynamic TerraLink achieves reliability through several independent mechanisms.

Mechanism	Purpose
Sequence Number	Packet identification
Duplicate Cache	Prevent repeated forwarding

TTL	Prevent loops
ACK	Confirm communication
Retransmission	Recover temporary packet loss
Route Table	Avoid unnecessary route discovery
RREQ	Discover unavailable routes
RREP	Establish usable route
RERR	Indicate failed route
Alternative Route	Continue communication after failure
Self-Healing	Recover dynamic topology

No single mechanism is responsible for reliability. The protocol uses them together.

Dynamic Self-Healing Principle

The defining feature of Dynamic TerraLink is:

The network does not depend on a permanently fixed communication path.

For example, assume:

Node 101

↓

Relay 1

↓

Relay 2

↓

Relay 3

↓

Base

If Relay 2 fails:

Node 101

↓

Relay 1

X

Relay 2

↓

Relay 3

↓

Base

the network detects the failed forwarding path.

If another relay is reachable:

Node 101

↓

Relay 1

↓

Relay 4

↓

Relay 3

↓

Base

communication can continue.

If no valid route exists, route discovery is initiated.

This provides the **dynamic and self-healing behaviour** of TerraLink.

29. End-to-End SOS Example

Consider:

Victim Node = 101

Relay Nodes = 1, 2, 3

Base Node = 4

Normal operation:

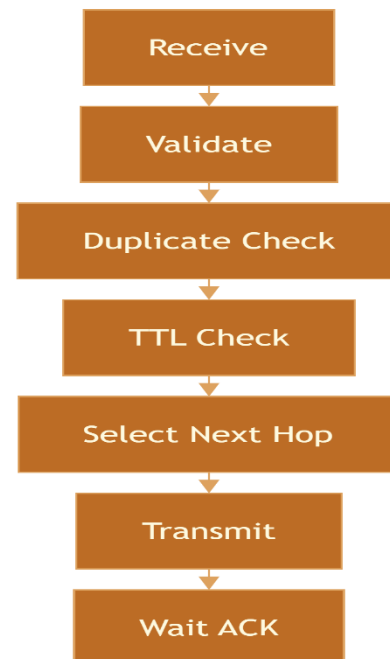
101 → 1 → 2 → 3 → 4

Finally:

Relay 3 → Base 4

The base registers the SOS.

At each relay:



31. Protocol Design Principles

Dynamic TerraLink is designed around five fundamental principles:

1. Distributed Routing

No single relay maintains complete control of the network.

2. Local Decision Making

Each relay determines the best available next hop from its current routing information.

3. Controlled Forwarding

Duplicate filtering and TTL prevent uncontrolled packet propagation.

4. Reliable Delivery

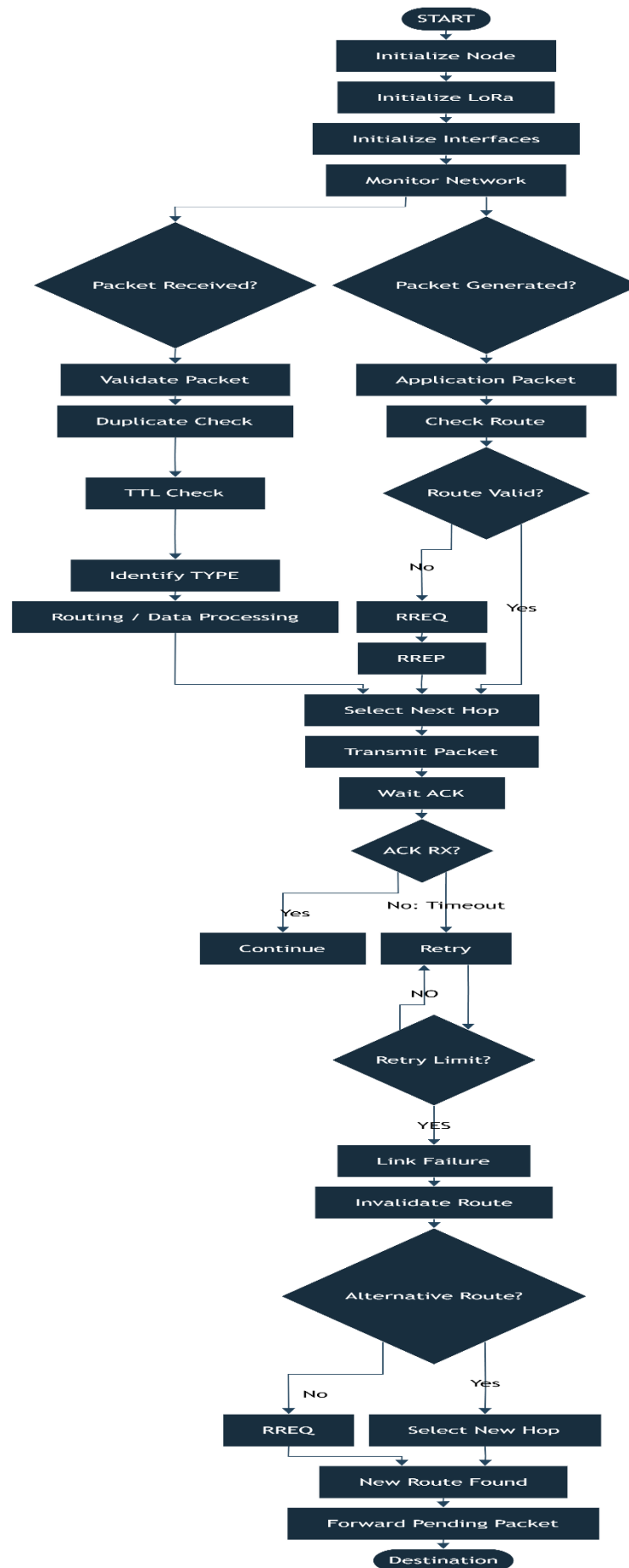
ACK and limited retransmission provide protection against temporary communication loss.

5. Self-Healing

Failed routes are detected and replaced through alternative-route selection or route discovery.

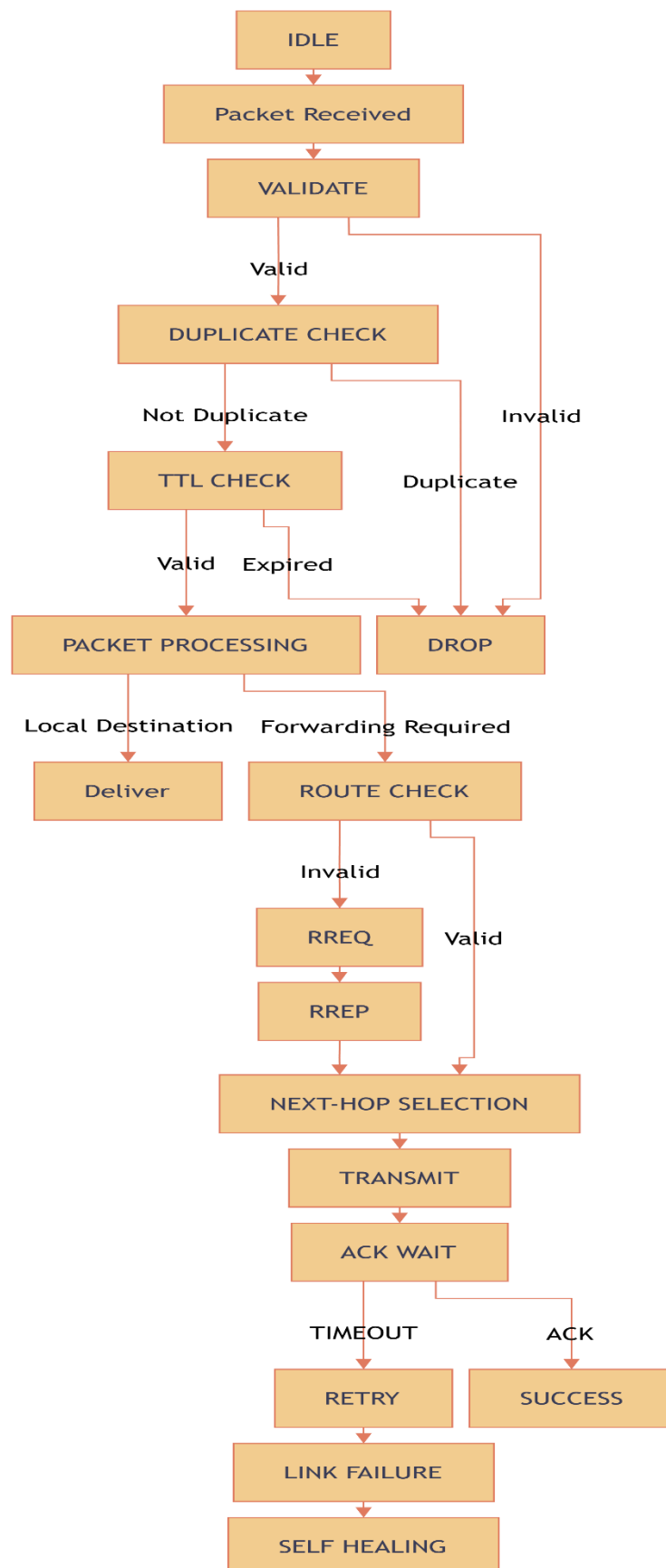
Complete Dynamic TerraLink Communication Algorithm

The latest overall algorithm can be represented as:



26. Dynamic TerraLink State Model

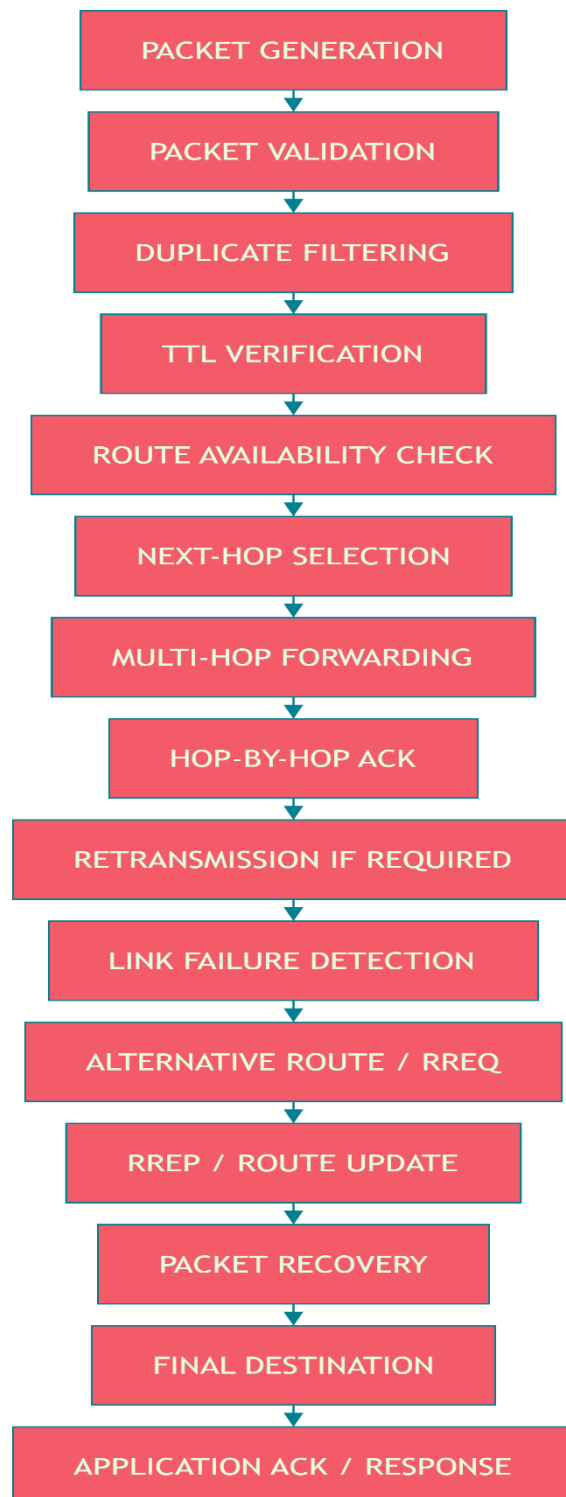
The relay node can be conceptually represented using the following states:



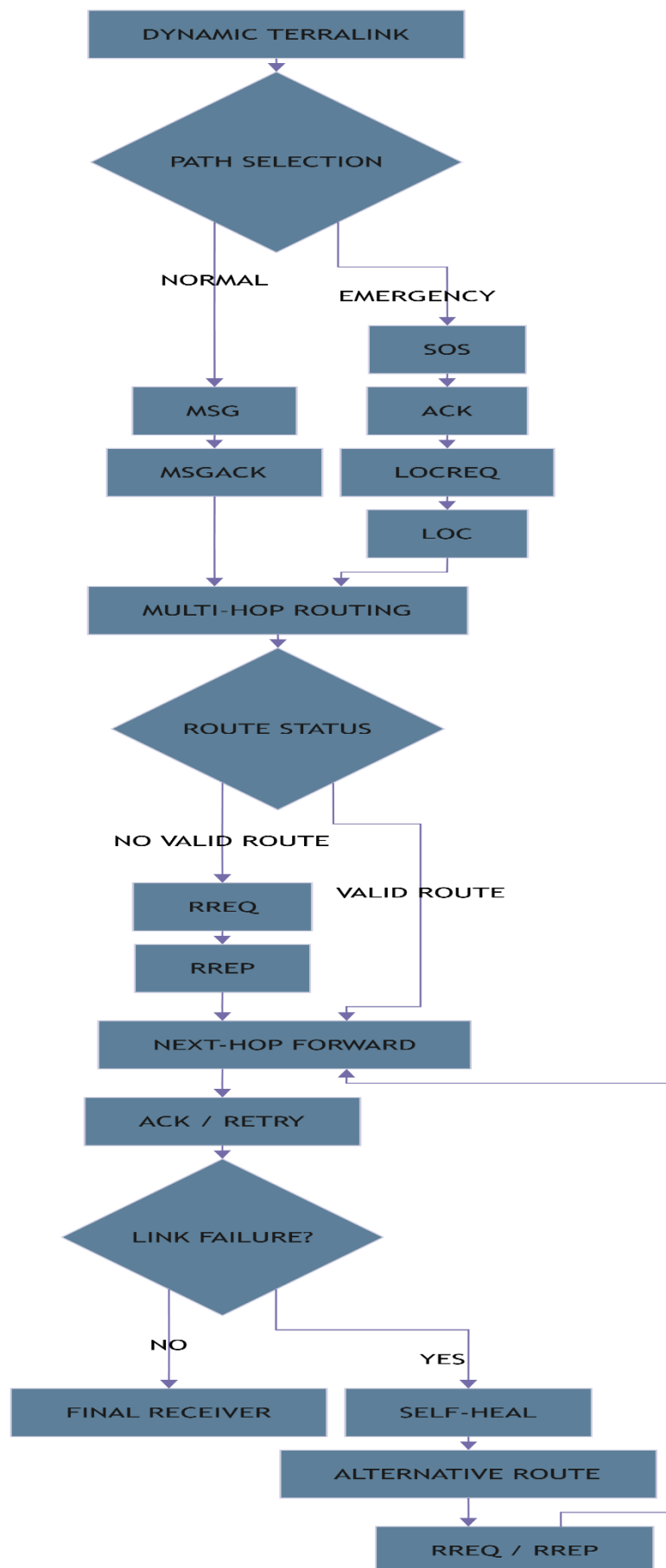
Final Protocol Summary

The **Dynamic TerraLink Communication Protocol** combines LoRa-based multi-hop communication with lightweight routing and reliability mechanisms suitable for resource-constrained embedded nodes.

Its complete communication process is:



The protocol supports both emergency and normal communication:



What changed from your previous document:

Previous documentation	Latest Dynamic TerraLink
Source-centric route discovery	Relay-centric routing
Mainly SOS communication	SOS + MSG + LOCREQ + LOC + acknowledgements
Generic forwarding	Explicit next-hop selection
Basic self-healing	Route invalidation + alternative route + RREQ/RREP recovery
SRC + SEQ duplicate filtering	Retained and explicitly integrated into packet processing
TTL	Retained
Hop-by-hop ACK	Retained and integrated with retry/link failure detection
AODV-inspired routing	Retained, but clearly stated as lightweight/inspired rather than full AODV
Static emergency path emphasis	Dynamic topology / self-healing emphasis
End node participates in routing conceptually	End node generates/consumes; relay nodes route
SOS-focused	Bidirectional emergency communication
Basic packet fields	TYPE, PID, SRC, DST, SEQ, TTL, HOP + application payloads